sector_balanced_max_legacy

equity strategy | family: equity-max | status: reference | derived from `sector_balanced_max`

What this strategy does

sector_balanced with bigger bets for more return: base unit 0.012, 6x leverage cap -- ORIGINAL leverage cap (new positions only, open book ignored)

Why it should work. Same drift, with sector and size risk diversified away so only the earnings signal is left.

Design notes (from the strategy's source file).

Max-size variants ("<name>_max") of the hand-built equity strategies.

Same signal, trim, tilt and neutralization as each base strategy; only the two dials that set how big and how levered the book is change (base_unit_fraction, gross-leverage cap = 6x). The multiplier for each was picked by sweeping 1.5x..12x the original base unit and taking the highest return subject to Sharpe >= 0.6 and max drawdown >= -40% (a worse drawdown is a fund-ending event, not a risk preference). The untouched 5%-of-ADV liquidity cap remains the real capacity constraint, and bites roughly four times as often at this sizing.

These are generated from the base strategy classes, so a change to a base strategy's weights flows through automatically.

parameter	value
base_unit_fraction	0.012
max_gross_leverage	6.0
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	priority
exposure_cap	entry_cohort
beta_overlay	0.0

Performance

metric	sector balanced max legacy	market (CRSP VW)
Annualized return	12.15%	9.94%
Annualized volatility	12.20%	19.35%
Sharpe (NAV-pct)	1.00	0.59
Sortino	1.45	
Max drawdown	-36.5%	-55.5%
Calmar	0.33	
Final NAV ($10M start)	$93,270,642	
Worst day	-7.14%	
% positive days	51.5%	

Sharpe 1996-2001	1.53	
Sharpe 2002-2007	1.71	
Sharpe 2008-2013	-0.07	

Correlation of daily returns with the market: **-0.02**.

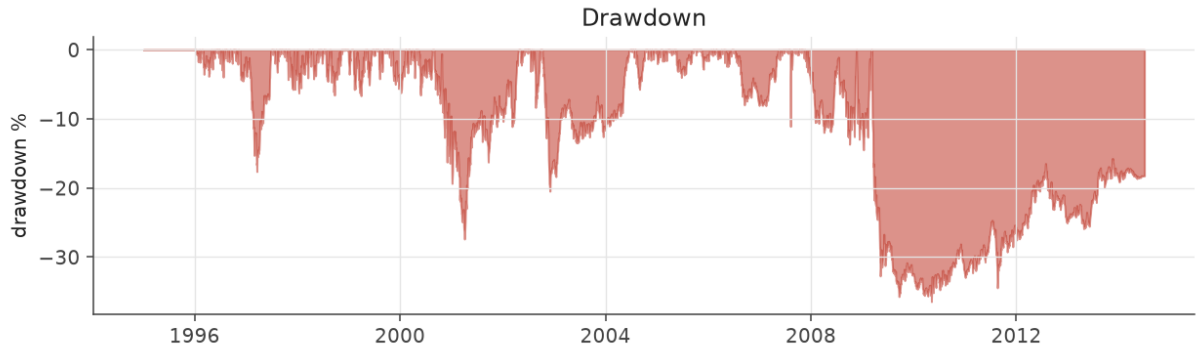
• best Sharpe among equity: `calm_rank_tilt` (1.47); this one: 1

• best return among equity: `market_boost_max` (0.116); this one: 0.121

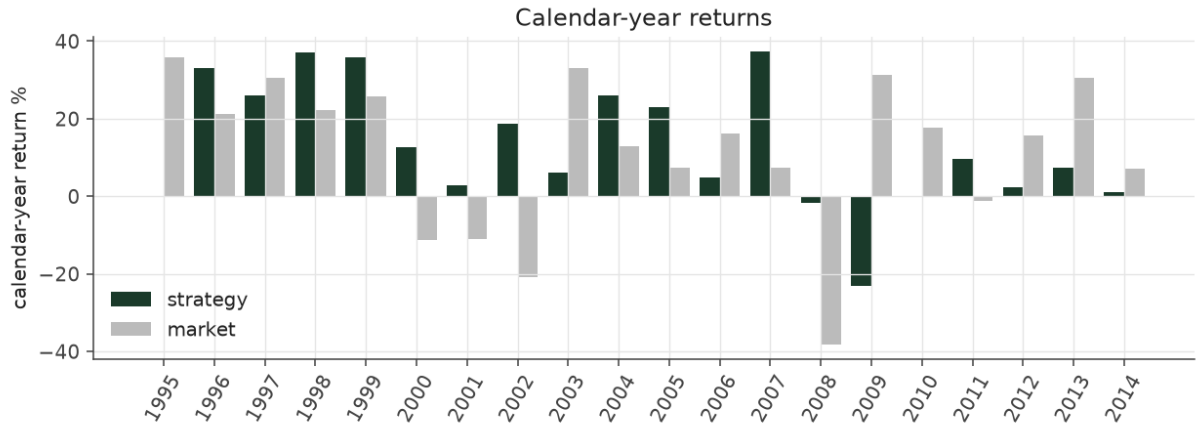
• best Calmar among equity: `calm_rank_tilt` (0.442); this one: 0.333



NAV (log scale)



Drawdown from running peak

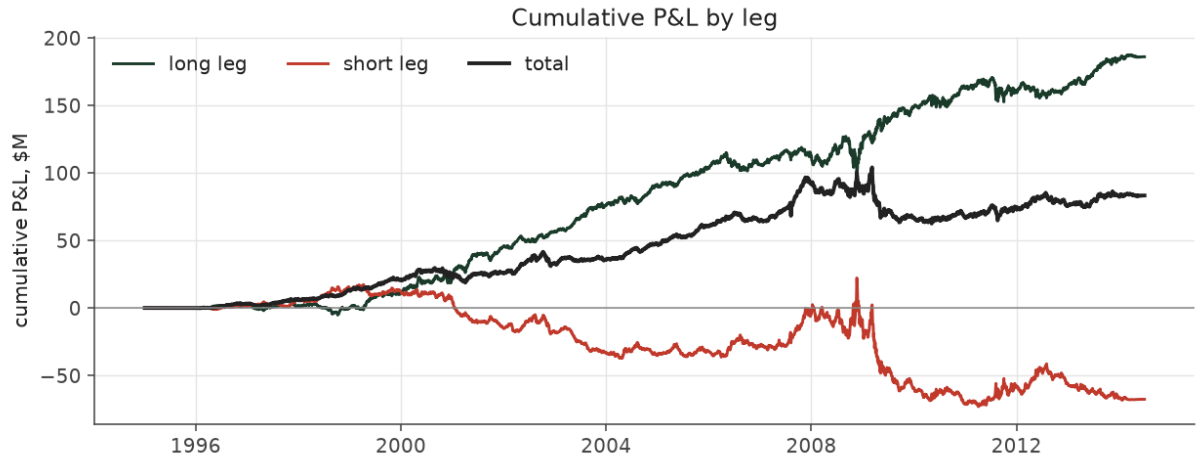


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	185.98	-67.62
P&L after costs ($M)	168.87	-85.60
of which common (average-event) component ($M)	175.41	-173.00

of which excess / PEAD component ($M)	9.52	105.97
excess minus costs ($M)	-7.56	88.03
gross return per $ traded (bp)	166.77	-59.88
positions in the money (hit rate)	0.56	0.49
stand-alone Sharpe (leg only)	1.06	-0.22

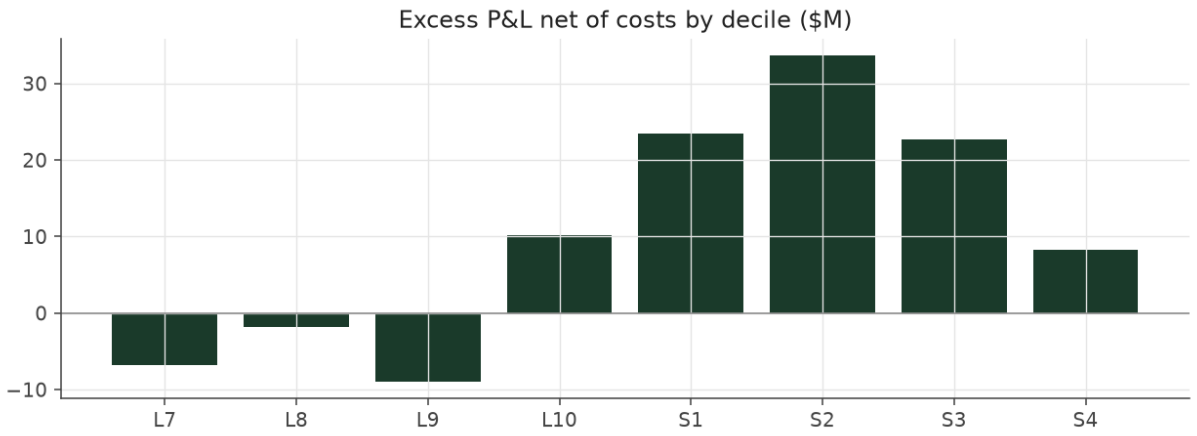
Daily P&L correlation between the two legs: **-0.77**.

• **The short leg loses money in absolute terms** (-67.6M before costs) while the long leg earns +186.0M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 175.4M / 173.0M. Stripping that common component, the surprise signal itself earned +9.5M on the long side and +106.0M on the short side.

• The two legs' daily P&L are -0.77 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 1.06 long / -0.22 short vs 1.00 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	7	7,718	1,052.78	11.28	-5.18	1.67	-6.85	-8.5%	-65.07
long	8	16,225	2,721.85	42.43	2.34	4.23	-1.89	-2.3%	-6.95
long	9	17,480	3,520.62	43.56	-3.69	5.32	-9.01	-11.2%	-25.59
long	10	16,939	3,857.22	71.60	16.06	5.90	10.16	12.6%	26.33
short	1	16,755	3,662.36	-27.10	29.32	5.92	23.40	29.1%	63.89
short	2	16,916	3,529.55	-24.03	39.18	5.54	33.64	41.8%	95.32
short	3	15,579	2,858.54	-23.05	27.22	4.57	22.65	28.1%	79.22
short	4	9,015	1,241.23	-11.41	10.25	1.95	8.30	10.3%	66.91



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	12,943	580.43	4.74	8.80	1.91	6.89	8.6%	118.74
2	22,307	2,425.20	15.93	23.41	5.94	17.47	21.7%	72.03

3	25,252	4,605.79	25.85	32.96	8.66	24.31	30.2%	52.78
4	27,425	6,685.72	17.20	24.36	9.76	14.60	18.1%	21.84
5	28,700	8,147.02	19.55	25.97	8.84	17.13	21.3%	21.03

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	11,934	272.18	5.63	7.61	1.36	6.25	7.8%	229.46
2	22,802	1,937.75	12.15	20.53	5.81	14.71	18.3%	75.93
3	25,740	4,782.62	26.45	37.55	9.56	27.99	34.8%	58.52
4	27,494	7,253.58	21.14	28.67	10.15	18.51	23.0%	25.52
5	28,652	8,196.33	17.81	21.03	8.20	12.83	15.9%	15.65

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	45,165	9,600.44	-14.15	35.15	14.06	21.09	26.2%	21.97
41-60d	25,072	4,938.28	27.47	18.39	7.84	10.56	13.1%	21.38
61-90d	34,592	6,224.84	23.02	26.24	10.15	16.09	20.0%	25.84
>90d	11,798	1,680.60	46.93	35.71	3.04	32.67	40.6%	194.40

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	32,056	3,008.84	26.65	30.08	4.72	25.36	31.5%	84.28
2002-07	39,668	7,889.54	61.23	69.53	12.19	57.34	71.3%	72.68
2008-13	44,903	11,545.77	-4.60	15.88	18.18	-2.30	-2.9%	-1.99

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	21,429	2,583.01	6.03	8.57	4.33	4.24	5.3%	16.40
Chems	2,571	1,235.93	7.33	8.92	1.67	7.26	9.0%	58.73
Durbl	2,800	1,290.38	2.69	4.58	1.78	2.81	3.5%	21.73
Enrgy	6,106	1,767.37	16.94	20.27	2.50	17.77	22.1%	100.55
Hlth	11,413	2,280.39	12.88	14.71	3.68	11.03	13.7%	48.37
Manuf	11,825	2,139.00	5.43	8.60	3.40	5.20	6.5%	24.31
Money	17,141	2,149.44	11.87	16.66	3.75	12.91	16.0%	60.07
NoDur	5,000	1,595.65	5.49	6.46	2.31	4.15	5.2%	26.03
Other	18,022	2,447.24	8.16	13.06	4.30	8.76	10.9%	35.78
Shops	11,767	2,235.81	13.10	15.48	3.56	11.93	14.8%	53.34

Telcm	3,932	1,386.74	-7.51	-4.84	1.87	-6.71	-8.3%	-48.41
Utils	4,621	1,333.20	0.86	3.03	1.96	1.06	1.3%	7.98

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	38,814	8,085.79	2.77	19.33	11.77	7.57	9.4%	9.36
low vol	39,075	6,848.95	38.30	46.76	11.52	35.24	43.8%	51.45
mid vol	37,173	7,280.81	41.91	49.66	11.41	38.26	47.5%	52.54

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	12.15%	-0.00%
0.5%	10.7	11.45%	0.70%
1.0%	21.5	10.65%	1.50%
2.0%	43.0	8.65%	3.50%
5.0%	107.4	-100.00%	112.15%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.